

# Unraveling the Kagome Antiferromagnetic $3J$ Model and Its Materials: An Integrated Approach

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We investigate the ground-state and finite-temperature properties of the kagome antiferromagnetic Heisenberg model with three inequivalent couplings, dubbed the  $3J$  model, which is designed for the candidate Dirac quantum spin liquid (QSL) material  $\text{YCu}_3(\text{OH})_6\text{Br}_2[\text{Br}_{1-x}(\text{OH})_x]$  [1]. Employing large-scale density-matrix renormalization group (DMRG) supplemented by neural quantum states (NQS) simulations, we identify an intermediate QSL phase between two magnetically ordered phases. We also find that this QSL is separated from the kagome spin liquid ground state at the isotropic limit. To establish a direct comparison with experiments, we compute the specific heat of the model by means of advanced exponential (XTRG) and tangent-space (tanTRG) thermal tensor-network methods. In the magnetically ordered phase, the specific heat over temperature exhibits a shoulder at a temperature that is a fraction of the coupling strength  $J_\square$ , which disappears in the QSL phase. These universal behaviors are consistent with the experimentally observed specific heat in  $3J$  materials for both ordered and QSL candidate samples. Our work thus connects microscopic models with experimentally measurable signatures, exemplifying an integrated approach to understanding QSL phenomena in frustrated quantum magnets [2], with  $3J$  materials serving as a representative case and providing a foundation for future studies.

**Introduction.**— Quantum magnets provide a fertile ground for exploring exotic quantum phases, among which quantum spin liquids (QSLs) [2–7] are particularly intriguing and elusive due to their departure from conventional ordered states. Rather than developing conventional long-range order, QSLs can host long-range quantum entanglement, fractionalized excitations, and emergent gauge structures, providing a natural setting for quantum phases beyond the Landau paradigm of symmetry breaking. One promising platform for realizing QSLs is the highly frustrated spin-1/2 kagome antiferromagnetic Heisenberg model [8–15]. Despite extensive numerical and theoretical efforts, however, the nature of its ground state remains debated, with various studies supporting competing scenarios, e.g., a gapped QSL [16–24], a gapless QSL [25–32] or valence-bond crystal (VBC) states [33–36]. This theoretical ambiguity is compounded by experimental challenges: disorder and magnetic impurities in candidate kagome materials [37–49] can mask intrinsic signatures of QSLs in thermodynamic and spectroscopic measurements and, in some cases, may destabilize the underlying quantum phase.

Recent experimental advances in kagome antiferromagnetic compounds  $\text{YCu}_3(\text{OH})_{6+x}\text{X}_{3-x}$  ( $\text{X}=\text{Cl}$  or  $\text{Br}$ ) [1, 50–59] have opened a promising avenue toward realizing a Dirac QSL with gapless Dirac excitations [3, 4]. Rather than forming an isotropic kagome lattice, these materials exhibit a weakly distorted kagome geometry with three inequivalent nearest-neighbor couplings,  $J_1$ ,  $J_2$ , and  $J_\square$ , defining the so-called  $3J$  model, as illustrated in the inset of Fig. 1. This distortion originates from the displacement of  $\text{Y}^{3+}$  ions out of the kagome plane by the polar  $(\text{OH})^-$  ligand, which splits the uni-

form hexagon coupling into two alternating couplings,  $J_1$  and  $J_2$  [52, 53]. Experiments on candidate  $3J$  materials have revealed the  $(1/3, 1/3)$  ordered phase [57, 58] and provided evidence for a possible Dirac QSL from specific-heat [51, 56, 57] and inelastic neutron scattering [1, 54] measurements. Despite these experimental indications, theoretical understanding of the  $3J$  model remains at an early stage. While the classical phase diagram has been established [60] and recent simulations have provided only a preliminary characterization of the excitation spectrum, with agreement with experiment largely limited to the high-energy regime [54]. More fundamentally, unbiased numerical simulations of the phase diagram of the  $3J$  model, particularly the nature and stability of the putative QSL, is still largely absent. This leaves several essential questions unresolved: Does the QSL survive in the presence of the bond anisotropy intrinsic to the  $3J$  model? How does it relate to the kagome spin liquid (KSL) of the isotropic ( $1J$ ) model? Do they represent the same quantum phase, or does the bond anisotropy stabilize a distinct QSL? How does it evolve into the neighboring  $(1/3, 1/3)$  ordered phases? Moreover, a systematic understanding of its finite-temperature properties and their connection to experimental observations remains elusive. Addressing these questions is essential for establishing whether the  $3J$  model can provide a robust theoretical framework for the experimentally observed QSL phenomena.

Motivated by the integrated strategy [2], in this Letter we combine four state-of-the-art quantum many-body computational approaches with measurements on multiple experimental samples to investigate the kagome antiferromagnetic  $3J$  model, providing a concrete demonstration of the strength of

such an integrated strategy. Rather than relying on any single numerical method or experimental signature, we establish the ground-state phase diagram via complementary large-scale density-matrix renormalization group (DMRG) [61–64] and neural quantum states (NQS) [65–68] calculations, characterize the finite-temperature behavior using advanced exponential (XTRG) [69] and tangent-space (tanTRG) [70] thermal tensor-network methods, and compare the resulting numerical predictions against specific-heat measurements from multiple samples, represented by the ordered  $\text{Y}_3\text{Cu}_9(\text{OH})_{19}\text{Cl}_8$  [57] and QSL candidate materials  $\text{YCu}_3(\text{OH})_6\text{Br}_2[\text{Br}_{1-x}(\text{OH})_x]$  [1, 51, 53] and  $\text{LuCu}_3(\text{OH})_6\text{Br}_2[\text{Br}_x(\text{OH})_{1-x}]$  [56] samples.

This combined strategy not only allows us to exploit the respective strengths of different numerical approaches using experiment as an independent touchstone, but also enables us to cross-validate and synthesize their insights to uncover the emergent physics of the  $3J$  model beyond the reach of any single method. In particular, it provides an unbiased route to establishing the quantum phase diagram, suggests a gapless intermediate QSL phase, and clarifies the connection of this phase to neighboring ordered phases and to the experimentally observed finite-temperature behavior.

**Model and Method.**– The Hamiltonian of kagome antiferromagnetic  $3J$  model is

$$H = J_1 \sum_{\langle i,j \rangle_1} \mathbf{S}_i \cdot \mathbf{S}_j + J_2 \sum_{\langle i,j \rangle_2} \mathbf{S}_i \cdot \mathbf{S}_j + J_\square \sum_{\langle i,j \rangle_\square} \mathbf{S}_i \cdot \mathbf{S}_j, \quad (1)$$

where  $\mathbf{S}_i$  denotes the spin-1/2 operator at site  $i$  (white dots in the inset of Fig. 1). All three couplings are antiferromagnetic,  $J_1, J_2, J_\square > 0$ , and correspond to nearest-neighbor interactions. To determine its ground-state phase diagram, we employ complementary approaches based on DMRG and NQS. We set  $J_\square = 1$  as the energy unit and focus on the parameter regime  $J_1, J_2 \in [0, 1]$ . For the DMRG calculations, we consider YC cylindrical geometries with a fixed circumference of  $L_y = 3$  (YC6) and system lengths  $L_x = 6, 12$  and  $24$ , compatible with the 27-site magnetic unit cell of the  $3J$  model. These correspond to total spin numbers  $N = 3 \times L_x \times L_y = 54, 108$  and  $216$ , respectively. We keep the maximum bond dimension up to  $D = 15,000$  with  $U(1)$  symmetry implemented, achieving well-converged results with truncation errors below  $1.0 \times 10^{-8}$ . For the NQS calculations, we adopt a Vision Transformer (ViT) architecture with factored attention [68, 71, 72], implemented in two variants that differ in their enforcement of translation symmetry. These simulations are performed on an  $N = 3 \times 6 \times 6$  system with periodic boundary conditions, and the network parameters are optimized by variational Monte Carlo [73]. For finite-temperature properties, we employ the advanced tensor-network methods XTRG and tanTRG as complementary approaches for cross-validation. Both methods are applied to YC6 cylinders with a maximum bond dimension of  $D = 2000$ , which yields well-converged results. Further technical details of the algorithms are provided in the Supplemental Material (SM) [74].

**Phase Diagram.**– Our ground-state results for the  $3J$  model obtained from DMRG and NQS calculations are summarized

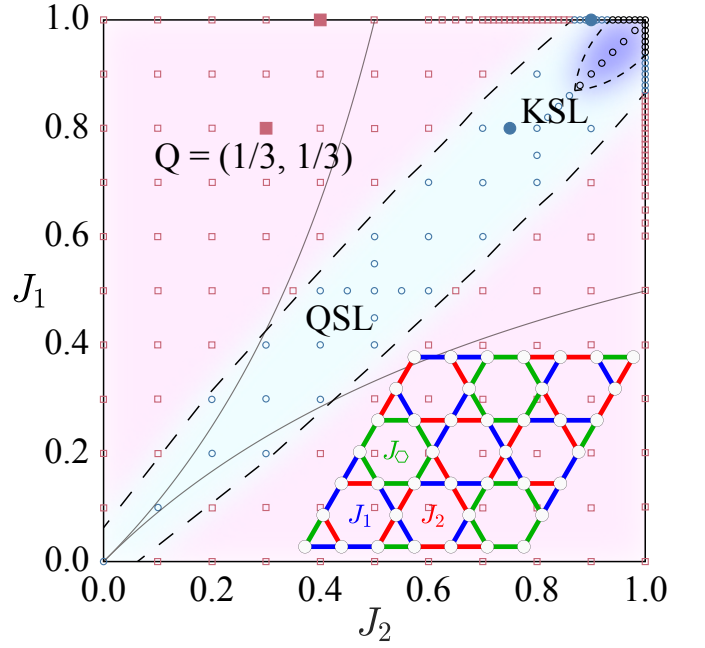


FIG. 1. **Ground-state phase diagram of the  $3J$  model.** Two  $(1/3, 1/3)$  ordered phases and an intermediate QSL phase are identified within  $0 \leq J_{1,2} \leq 1$ . The QSL phase is separated from the KSL ground state around the isotropic limit,  $J_1 = J_2 = J_\square$ . Open symbols denote the calculated parameter points, while solid symbols indicate the parameter points used for specific-heat calculations. The black dashed lines indicate the quantum phase boundaries determined in this work, and the gray solid lines show the classical phase boundaries obtained from Ref. [60] for comparison. The inset displays an enlarged 27-site unit cell (in the  $(1/3, 1/3)$  ordered phase) of the  $3J$  model on the kagome lattice, with different bond colors indicating the distinct Heisenberg interactions defined in Eq. (1).

in the phase diagram Fig. 1, where  $J_1$  and  $J_2$  are varied with  $J_\square = 1$  fixed. We identify a candidate QSL phase (light-blue region) sandwiched between two  $(1/3, 1/3)$  ordered phases [60] (light-pink regions). Compared with the classical phase boundaries (gray solid lines) determined by classical Monte Carlo calculations [60], the QSL region is reduced in the large- $J_{1,2}$  regimes due to enhanced quantum fluctuations. Moreover, it is concentrated around the diagonal symmetry axis  $J_1 = J_2$ , where the competing interactions are more balanced in strength, favoring the emergence of the QSL. Notably, we have evidence that this QSL phase is separated from the KSL ground state (light-purple regime) in the vicinity of the isotropic  $1J$  model by a phase transition. These results imply that distorted kagome  $3J$  materials are likely to host two distinct QSL phases.

We characterize these distinct quantum phases by examining the static spin structure factor  $S(\mathbf{k}) = (1/N) \sum_{i,j} \langle \mathbf{S}_i \cdot \mathbf{S}_j \rangle e^{i\mathbf{k} \cdot (\mathbf{r}_i - \mathbf{r}_j)}$ . In the  $(1/3, 1/3)$  ordered phases,  $S(\mathbf{k})$  show Bragg peaks at either the  $\mathbf{K}_1$  points [cf. Fig. 2(a)] or the  $\mathbf{K}_2$  points [cf. Fig. 2(b)]. It is worth pointing out that these two  $(1/3, 1/3)$  ordered phases are physically equivalent, as both correspond to the same ordering wave vector  $\mathbf{Q} = (1/3, 1/3)$ .

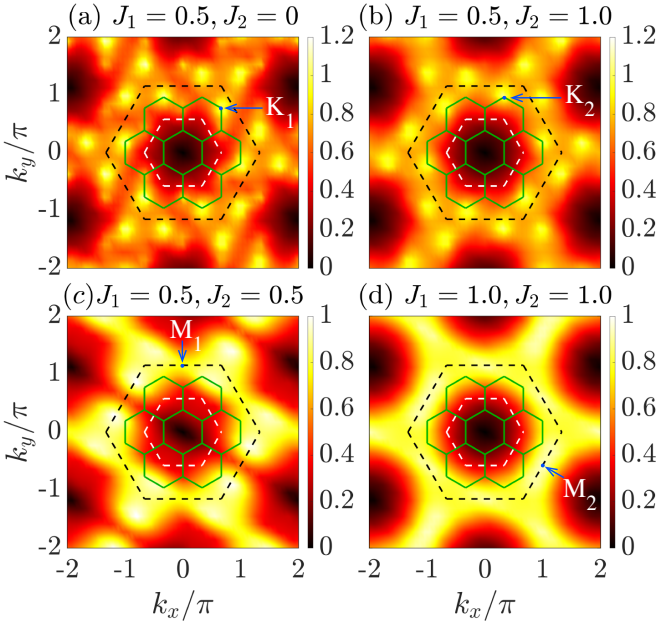


FIG. 2. **Static spin structure factors.** Representative static spin structure factors  $S(\mathbf{k})$  in the (a,b)  $(1/3, 1/3)$  ordered phases, (c) QSL phase, and (d) KSL phase. The ordering peaks in (a) occur at  $\mathbf{Q} = \mathbf{K}_1$ , whereas those in (b) occur at  $\mathbf{Q} = \mathbf{K}_2$ . The positions of  $\mathbf{M}_1$  and  $\mathbf{M}_2$  are also denoted directly in (c) and (d). The black dashed hexagon denotes the Brillouin zone (BZ) of the triangular lattice (1-site unit cell), the white dashed hexagon represents the BZ of the kagome lattice (3-site unit cell), and the green solid hexagon corresponds to the BZ of the  $3J$  model (9-site unit cell). Here, all  $S(\mathbf{k})$  are obtained by Fourier transformation of all-to-all spin-spin correlations. The corresponding NQS results on the  $6 \times 6$  torus are illustrated in Fig. S2 of the SM [74]

The apparent distinction between  $\mathbf{K}_1$  and  $\mathbf{K}_2$  arises from a transformation that effectively swaps  $J_1$  and  $J_2$  in the Hamiltonian. In stark contrast, the QSL phase exhibits no sharp Bragg peaks in  $S(\mathbf{k})$  [cf. Fig. 2(c)], but rather a broad and featureless distribution, reflecting the absence of long-range magnetic order. For comparison, we also present  $S(\mathbf{k})$  for the KSL phase in Fig. 2(d), which likewise exhibits disordered behavior. However, the topology of  $S(\mathbf{k})$  in these two spin liquid phases differs significantly:  $S(\mathbf{k})$  is discontinuous at  $\mathbf{M}_2$  points in the QSL phase, whereas it remains continuous at both  $\mathbf{M}_1$  and  $\mathbf{M}_2$  points in the KSL phase. We have verified this distinction across all parameter points within both the QSL and KSL regimes. On the other hand, the real-space spin correlations provide another clear distinction between the QSL and KSL phases [cf. Fig. S3 in the SM [74]]. In the QSL phase, the spin correlations decay algebraically as a power law, consistent with the behavior expected for a Dirac QSL state, whereas those in the KSL phase decay exponentially. Here, we refrain from drawing conclusions about whether the KSL phase is gapped or gapless in the two-dimensional limit, as our results are based solely on finite-size simulations on  $L_y = 3$  cylinders and an  $L_y = 6$  torus, while determining the nature of the KSL lies beyond the scope of the present

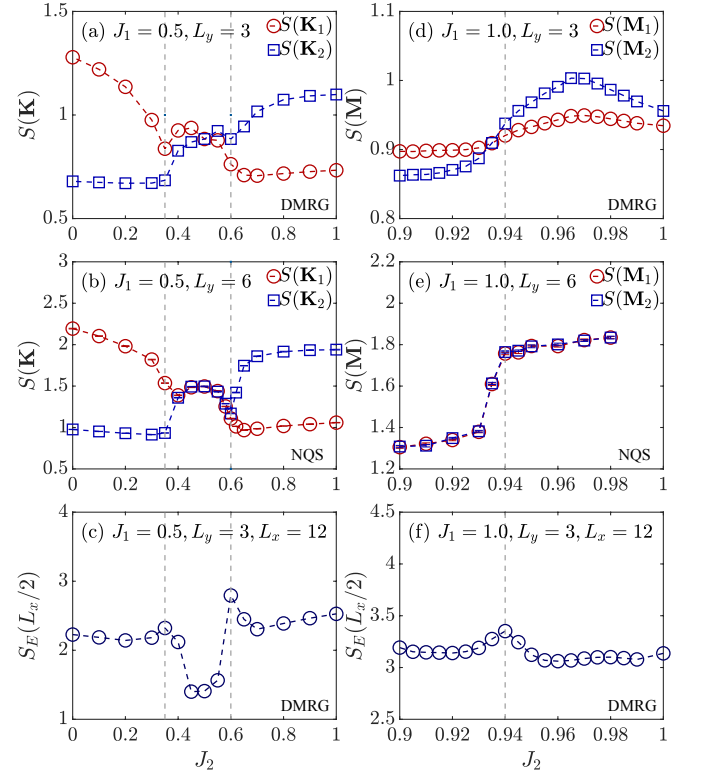


FIG. 3.  **$J_2$  dependence of  $S(\mathbf{k})$  and entanglement entropy.** (a)  $J_2$  dependence of  $S(\mathbf{k})$  at  $\mathbf{K}_1$  and  $\mathbf{K}_2$  for  $J_1 = 0.5$  on YC6 ( $L_y = 3, L_x = 12$ ) cylinders obtained by DMRG. The kink-like features near  $J_2 \approx 0.35$  and  $0.6$ , indicated by the gray dashed lines, signal phase transitions. (b) Same as (a) but on a torus ( $L_y = 6, L_x = 6$ ) obtained from NQS calculations. (c)  $J_2$  dependence of the entanglement entropy  $S_E(x)$  at the bulk of cylinders  $x = L_x/2$ , yielding critical points consistent with those identified in (a) and (b). (d-f) Same as (a-c), respectively, but for  $J_1 = 1.0$ , with  $S(\mathbf{k})$  evaluated at  $\mathbf{M}_1$  and  $\mathbf{M}_2$ , where the phase transition occur near  $J_2 \approx 0.94$ .

work. Instead, these results provide an additional distinction between the QSL and KSL phases without presupposing the gap structure of the KSL.

To accurately locate the phase boundaries, we track the evolution of  $S(\mathbf{k})$  with  $J_1$  and  $J_2$  at the ordering momenta  $\mathbf{K}_1$  and  $\mathbf{K}_2$ . As a representative cut through the phase diagram, we fix  $J_1 = 0.5$  and vary  $J_2$ , thereby traversing the  $(1/3, 1/3)$  ordered and QSL phases [cf. Fig. 3(a)]. As  $J_2$  increases, the dominant  $S(\mathbf{K}_1)$  is gradually suppressed. Around  $J_2 \approx 0.35$ , it exhibits a pronounced kink, accompanied by the gradual development of  $S(\mathbf{K}_2)$ , signaling a phase transition. In the intermediate regime  $0.35 < J_2 < 0.6$ , neither  $S(\mathbf{K}_1)$  nor  $S(\mathbf{K}_2)$  dominates, and both magnetic correlations are substantially suppressed, supporting the presence of a QSL phase. Upon further increasing  $J_2$ ,  $S(\mathbf{K}_1)$  is completely suppressed near  $J_2 \approx 0.6$ , while  $S(\mathbf{K}_2)$  develops a pronounced kink and becomes the dominant magnetic ordering tendency. Since the computational cost of DMRG grows exponentially with cylinder width  $L_y$ , we use NQS as a complementary approach to access wider systems on a torus with  $L_y = 6$  [Fig. 3(b)]. Re-



markably, the locations of the transition points remain stable against finite-size effects and boundary conditions, further supporting the robustness of the identified phase boundaries. In Fig. 3(c), we also show the evolution of the entanglement entropy  $S_E(x) = -\text{Tr}[\rho_x \ln \rho_x]$  from DMRG at the middle of the cylinders  $x = L_x/2$ , where  $\rho_x$  is the reduced density matrix of the subsystem with rung number  $x$ . Two kinks are clearly observed, whose locations are also consistent with those identified from Figs. 3(a-b). Moreover, the entanglement entropy is larger in the ordered phase, consistent with the expected contribution from Goldstone modes [75]. Additionally, we characterize the QSL–KSL transition by tracking the evolution of  $S(\mathbf{k})$  at  $\mathbf{M}_1$  and  $\mathbf{M}_2$  along a cut with fixed  $J_1 = 1$ . Beyond the distinct  $S(\mathbf{k})$  profiles shown in Figs. 2(c,d), the QSL–KSL transition is signaled by pronounced kinks in  $S(\mathbf{M}_1)$  and  $S(\mathbf{M}_2)$  near  $J_2 \simeq 0.94$  [cf. Figs. 3(d,e)], accompanied by singular behavior in  $S_E(x)$  [cf. Fig. 3(f)]. As an additional cross-check of the phase boundaries, we examine the evolution of  $S(\mathbf{M})$  in the  $(1/3, 1/3)$  ordered phases and  $S(\mathbf{K})$  in the QSL and KSL phases [cf. Fig. S1 in the SM [74]]. The transition points extracted from these independent diagnostics are in excellent agreement.

Finally, to further examine whether the ground state may host VBC states, we also analyze the bond energies across the phase diagram. In both the QSL and KSL phases, we find no discernible signatures of lattice-symmetry breaking, further ruling out VBC as the underlying ground-state order in these regimes [cf. Fig. S4 in the SM [74]].

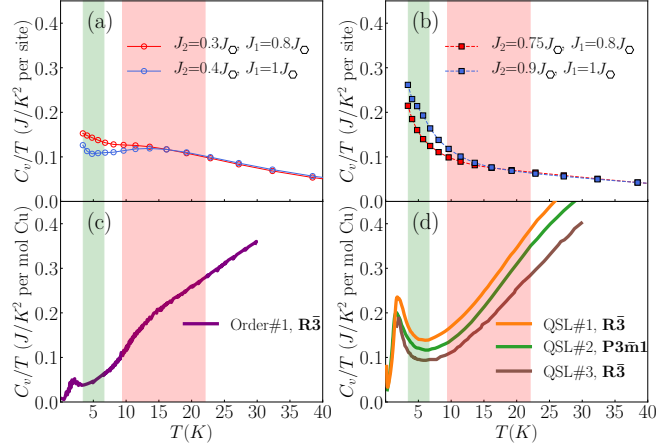


FIG. 4. **Computed and experimental specific heat.** (a) and (b) Calculated  $C_v/T$  from XTRG for representative parameter sets in the  $(1/3, 1/3)$  ordered and QSL phases, respectively. The temperature scale is set by  $J_\square = 63$  K following the experimental estimate [76]. (c) Experimental  $C_v/T$  for  $\text{Y}_3\text{Cu}_9(\text{OH})_{19}\text{Cl}_8$  from Ref. [57] (ordered, marked as Order#1), which crystallizes in the  $R\bar{3}$  space group and realizes the  $(1/3, 1/3)$ -ordered phase. (d) Experimental  $C_v/T$  for three candidate QSL materials: disordered  $\text{LuCu}_3(\text{OH})_6\text{Br}_2[\text{Br}_x(\text{OH})_{1-x}]$  in the  $R\bar{3}$  space group (orange, QSL#1) [56];  $\text{YCu}_3(\text{OH})_6\text{Br}_2[\text{Br}_x(\text{OH})_{1-x}]$  (green, QSL#2), which crystallizes in the  $P\bar{3}m1$  space group but is also considered a  $3J$ -model QSL candidate [51, 54]; and  $\text{YCu}_3(\text{OH})_6\text{O}_{1/3}\text{Cl}_{8/3}$  (brown, QSL#3) [77], which crystallizes in the  $R\bar{3}$  space group.

**Specific Heat.**— To establish a direct connection between the ground states and experimentally accessible thermal signatures, we calculate the specific heat using XTRG and tanTRG, and present the results as  $C_v/T$ . We consider four representative parameter sets spanning the  $(1/3, 1/3)$  ordered and QSL phases:  $(J_1, J_2) = (0.8, 0.3)J_\square$  and  $(1, 0.4)J_\square$  in the  $(1/3, 1/3)$  ordered phase;  $(0.8, 0.75)J_\square$  and  $(1, 0.85)J_\square$  in the QSL phase. The corresponding numerical results are shown in Figs. 4(a-b), with the temperature scale set by  $J_\square = 63$  K following the experimental estimate [76]. We compare these results with experimental data for the ordered material  $\text{Y}_3\text{Cu}_9(\text{OH})_{19}\text{Cl}_8$  [cf. Fig. 4(c)] and three candidate QSL materials, including  $\text{LuCu}_3(\text{OH})_6\text{Br}_2[\text{Br}_x(\text{OH})_{1-x}]$ ,  $\text{YCu}_3(\text{OH})_6\text{O}_{1/3}\text{Cl}_{8/3}$ , and  $\text{YCu}_3(\text{OH})_6\text{Br}_2[\text{Br}_x(\text{OH})_{1-x}]$  [cf. Fig. 4(d)]. We note that the experimental  $C_v/T$  data contain a nonmagnetic phonon contribution that is difficult to subtract reliably at low temperatures, especially below 25 K. Despite this limitation, the experimental data exhibit clear temperature-dependent features that allow a meaningful comparison with our numerical results, revealing two distinct thermal signatures that differentiate the  $(1/3, 1/3)$ -ordered and QSL phases.

First, the  $(1/3, 1/3)$  ordered phase exhibits a shoulder-like anomaly in  $C_v/T$  at  $T \sim 10$ –22 K (red shading), which is absent in the QSL phase. This shoulder consistently appears around  $T \sim 15$  K in the ordered compound  $\text{Y}_3\text{Cu}_9(\text{OH})_{19}\text{Cl}_8$ , but is absent in all three candidate QSL materials. This difference suggests that the shoulder serves as a signature of the zero-temperature magnetic order and thus provides a thermal diagnostic for distinguishing the two phases. Second, at  $T \sim 5$  K (green shading),  $C_v/T$  in the ordered phase exhibits a pronounced minimum, reaching  $\sim 0.1$ , while it remains larger ( $\sim 0.2$ ) in the QSL phase. The experimental data show the same trend, with a deep low-temperature dip in the ordered compound and enhanced  $C_v/T$  in the QSL candidates. This enhancement implies a larger low-energy density of states in the QSL phase, which naturally accounts for the stronger low-temperature contribution.

Moreover, in Fig. 5(a), we compare XTRG and tanTRG in the QSL phase at  $(J_1, J_2) = (1, 0.9)J_\square$  with bond dimensions up to  $D = 2000$  to assess the robustness of our numerical results. The XTRG results are well converged for  $D = 1000$  and 2000 over the entire temperature range  $T/J_\square \in [0.05, 10]$ . In contrast, tanTRG exhibits pronounced  $D$  dependence at low temperatures, converging to the XTRG results only as  $D$  increases. This confirms the different performance of XTRG and tanTRG and supports the reliability of the results in Figs. 4(a-b). On the other hand, we also compare our XTRG results at the isotropic  $1J$  kagome limit with SPINPACK data from Ref. [78] [cf. Fig. 5(b)]. The good agreement further validates our approach. Finally, to highlight the distinct thermal behavior, we compare  $C_v/N$  at  $(J_1, J_2) = (1.0, 0.9)J_\square$  with that of the  $1J$  model. The QSL-phase data exhibit a prominent shoulder around  $T/J_\square \sim 0.1$ , whereas the  $1J$  model decreases monotonically and smoothly over the same range. This striking difference corroborates our ground-state phase diagram

analysis, providing strong evidence that the QSL phase of the  $3J$  model is not adiabatically connected to the  $1J$  limit.

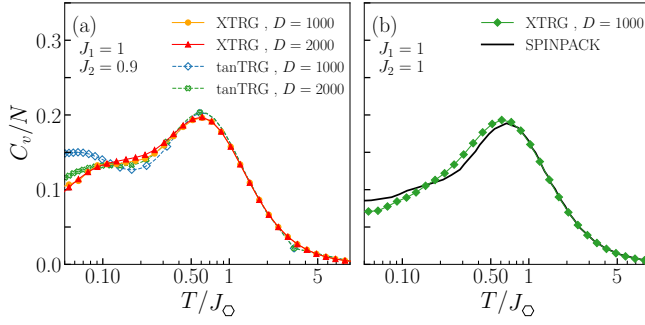


FIG. 5. **Specific heat of the  $3J$  and  $1J$  models.** (a) Calculated specific heat density  $C_v/N$ , from XTRG and tanTRG calculations with bond dimension  $D = 1000$  and  $2000$ . (b)  $C_v/N$  at the isotropic point  $J_1 = J_2 = J_O$ , from the XTRG calculations with  $D = 1000$ . The black line shows the corresponding  $C_v/N$  from SPINPACK calculations reported in Ref. [78].

**Summary and Discussion.**— In this work, we have established the ground-state and finite-temperature properties of the kagome antiferromagnetic  $3J$  model through an integrated investigation, combining two complementary ground-state methods (DMRG and NQS), two finite-temperature tensor-network approaches (XTRG and tanTRG), and experimental specific-heat data from four  $3J$  materials. These independent approaches consistently identify an intermediate QSL phase between two  $(1/3, 1/3)$  ordered phases and establish that it is distinct from the KSL phase at the isotropic  $1J$  limit. The consistency between DMRG and NQS results, despite their different geometries and effective system sizes, mitigates uncertainties associated with boundary conditions and finite-size effects, rendering the identification of the QSL phase particularly robust and compelling. The rich quantum phase diagram also highlights the intricate physics arising from the interplay of inequivalent exchange couplings. The calculated specific heat further captures the characteristic shoulder in the ordered regime and its disappearance in the QSL regime, in close agreement with experiments on both ordered and QSL samples.

Beyond establishing the phase diagram and finite-temperature signatures, our integrated analysis also raises several important questions concerning the low-energy physics of the QSL phase. Resolving the corresponding low-energy Dirac-cone structure in the excitation spectrum using both analytic [79] and numerical [80–83] approaches would provide a more direct characterization of its underlying field theory. On the experimental side, we note that our thermal measurements have not yet accessed a sufficiently low-temperature regime to resolve the  $T^2$  specific heat ( $C_v \propto T^2$ ) expected for a Dirac QSL [25, 84, 85]. Accessing this regime numerically will require further advances in finite-temperature numerical methods to enable reliable simulations at lower temperatures. Besides, the magnetic field response [84, 86, 87]

of the  $3J$  model also warrants investigation, including the intriguing  $1/9$  magnetization plateau and quantum oscillations observed therein [55, 76, 88–90], as well as possible field-induced phase transitions, which may provide further insight into the QSL-KSL transition and offer additional probes of the underlying low-energy theory.

In parallel, NQS approaches have demonstrated their ability to accurately capture ground states, low-lying excitations, and phase boundaries in frustrated magnets, including the square- and triangular-lattice  $J_1$ - $J_2$  models [91, 92] and the Shastry-Sutherland model [68, 93, 94]. Recent studies using variational wave functions based on the transformer architecture have reached periodic clusters of several hundred to more than a thousand spins on the square and triangular lattices [95, 96], enabling controlled finite-size extrapolations of energies and order parameters [96]. The  $3J$  model provides a natural yet demanding setting for such scaling efforts, owing to its enlarged unit cell, inequivalent exchange couplings and frustrated geometry, while its close connection to experimentally accessible materials makes quantitative theoretical predictions especially valuable. Extending NQS simulations to magnetization processes [97], excited-state spectra, and finite-temperature observables such as the specific heat [98] could therefore offer an independent route to the in-field, low-energy and thermodynamic properties of the  $3J$  model, complementary to conventional tensor-network approaches.

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**Data availability.**— The data are available from the authors upon reasonable request.

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## Supplemental Materials for: “Unraveling the Kagome Antiferromagnetic $3J$ Model and Its Materials: An Integrated Approach”

In the Supplemental Material, we provide additional results and methodological details supporting the conclusions presented in the main text. In Sec. I, we present additional ground-state results from DMRG and NQS calculations, including the evolution of the static spin structure factor, real-space spin correlations, and bond energies across the phase diagram. In Sec. II, we describe the algorithmic framework and implementation details of the NQS approach used to obtain the ground states. In Sec. III, we provide further technical details of the XTRG and tanTRG thermal tensor-network algorithms, together with the specific heat results for additional parameter points.

### I. ADDITIONAL GROUND-STATE RESULTS BY DMRG AND NQS

In the main text, we identify the phase transition points by tracking kink-like features in the evolution of  $S(\mathbf{K})$  in the  $(1/3, 1/3)$  ordered phases and  $S(\mathbf{M})$  in the QSL and KSL phases, together with singular behavior in the entanglement entropy. Here, we provide an additional diagnostic based on the static spin structure factor, using  $S(\mathbf{M})$  in the  $(1/3, 1/3)$  ordered phases and  $S(\mathbf{K})$  in the QSL and KSL phases to independently cross-check the transition points. In Fig. S1(a), we show the evolution of  $S(\mathbf{M}_1)$  and  $S(\mathbf{M}_2)$  as a function of  $J_2$  for  $J_1 = 0.5$ , obtained from DMRG calculations. Two pronounced kink-like features are observed near  $J_2 \approx 0.35$  and  $0.6$ , respectively, signaling changes in the underlying spin correlations and providing independent support for the phase boundaries shown in Figs. 3(a-c). For  $J_1 = 1.0$ , similar kink-like features emerge near  $J_2 \approx 0.86$  and  $0.94$  [cf. Fig. S1(b)]. The first kink, near  $J_2 \approx 0.86$ , marks the transition from the  $(1/3, 1/3)$  ordered phase to the QSL phase and is also observed in the evolution of  $S(\mathbf{M})$  [cf. Fig. S1(c)]. The second kink, near  $J_2 \approx 0.94$ , agrees excellently with the QSL–KSL transition point identified from the complementary diagnostics in Figs. 3(d-f), further supporting the robustness of the phase boundaries established from our ground-state calculations.

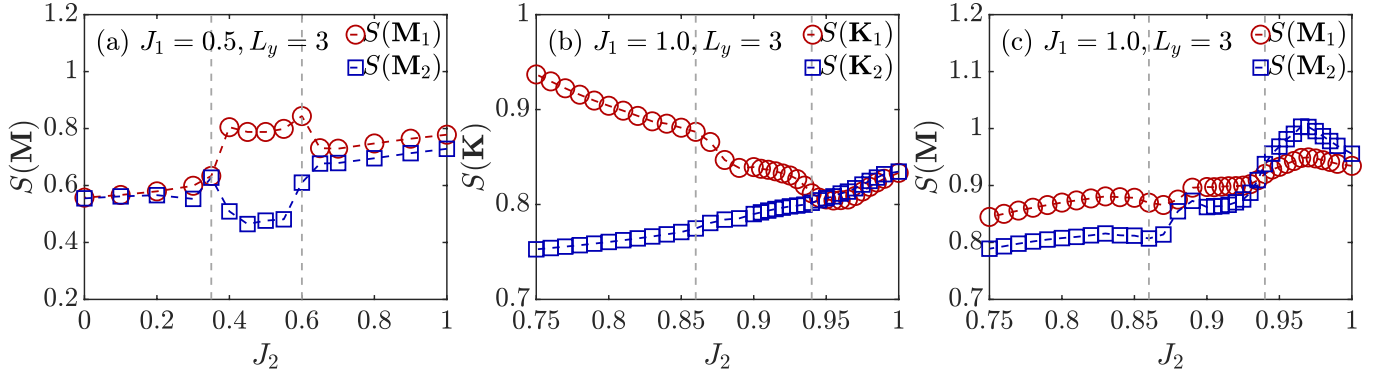


FIG. S1.  $J_2$  dependence of  $S(\mathbf{k})$  at the  $\mathbf{M}$  and  $\mathbf{K}$  points from DMRG calculations. (a)  $J_2$  dependence of  $S(\mathbf{k})$  at  $\mathbf{M}_1$  and  $\mathbf{M}_2$  for  $J_1 = 0.5$  on YC6 ( $L_y = 3$ ,  $L_x = 12$ ) cylinders. (b,c) Same as (a), but for  $J_1 = 1.0$ , with  $S(\mathbf{k})$  evaluated at the  $\mathbf{K}$  and  $\mathbf{M}$  points, respectively. The kink-like features are indicated by the gray dashed lines.

In Fig. S2, we also present the  $S(\mathbf{k})$  profiles obtained from NQS calculations, which are in excellent agreement with those shown in Fig. 2. One difference is that, in the QSL phase,  $S(\mathbf{k})$  remains continuous at  $\mathbf{M}_2$ , in contrast to the behavior observed in Fig. 2(c). This difference can be attributed to the distinct cylinder and torus geometries, with the latter restoring the  $C_3$  rotational symmetry. In addition to the distinct  $S(\mathbf{k})$  profiles shown in Figs. 2(c,d) and Figs. S2(c,d), the QSL and KSL phases can be further distinguished by their real-space spin correlations,  $F(r) = \langle \mathbf{S}_{x,y} \cdot \mathbf{S}_{x+r,y} \rangle$ , as presented in Fig. S3. In the QSL phase [cf. Figs. S3(a-f)],  $F(r)$  decays slowly and is well described by a power-law behavior, indicative of gapless excitations and consistent with the proposed Dirac QSL in the  $3J$  model. In contrast,  $F(r)$  in the KSL phase [cf. Figs. S3(g-i)] decays rapidly and is well fitted by an exponential behavior, with a very short correlation length  $\xi_s \sim 1$ . The distinct behavior observed for the QSL and KSL regimes on the finite YC6 cylinders provides an additional way to distinguish the two phases. Since the present results are restricted to finite-size systems, they do not by themselves determine whether the KSL is gapped or gapless in the two-dimensional limit.

In Fig. S4, we further examine the possibility of VBC order by calculating the nearest-neighbor bond energies. Figs. S4(a-b) show the spatial distribution of the bond energies on the YC6 kagome lattice in the QSL phase, with model parameters chosen consistently with the finite-temperature simulations. Within the present energy resolution, we find no evidence of broken lattice-translation symmetry, ruling out any discernible VBC order. Similarly, the bond energies in the KSL phase [cf. Fig. S4(d)] are

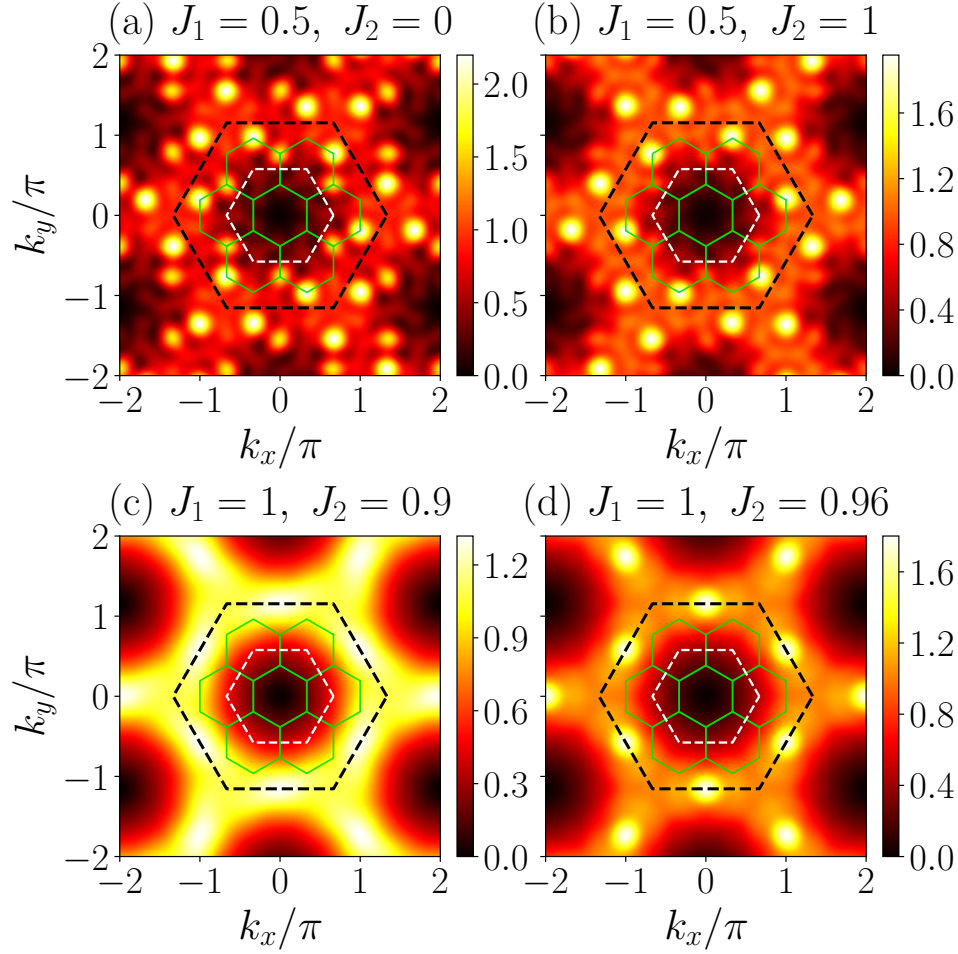


FIG. S2. **Static spin structure factors from NQS.** Representative static spin structure factors  $S(\mathbf{k})$  obtained from the NQS calculations on the  $6 \times 6 \times 3$  torus in the (a,b)  $(1/3, 1/3)$  ordered phases, (c) QSL phase, and (d) KSL phase, with couplings in units of  $J_\square$ . Pronounced peaks occur in (a,b) at the ordering vectors  $\mathbf{Q} = \mathbf{K}_1$  and  $\mathbf{Q} = \mathbf{K}_2$ , respectively, at the same positions as in Figs. 2(a,b). Brillouin zone conventions are as in Fig. 2. All maps are Fourier transforms of the all-to-all spin-spin correlations on the torus, estimated with  $2^{16}$  Monte Carlo samples and plotted as continuous functions of  $\mathbf{k}$ .

highly uniform, with no discernible signature of VBC order. It is worth noting that along the diagonal line  $J_1 = J_2$ , as  $J_1$  and  $J_2$  are gradually decrease, the dominant  $J_\square$  term leads to a characteristic spatial pattern in the bond energies, manifested as the  $J_\square$ -dominated hexagons [green hexagons in Fig. S4(c)]. This behavior is expected, since in the limit of  $J_1, J_2 \rightarrow 0$ , the system approaches a collection of decoupled hexagons. Importantly, however, this pattern does not break any symmetry of the  $3J$  model Hamiltonian. It merely reflects the modulation of bond energies within the unit cell and therefore does not constitute VBC order.

## II. NQS CALCULATION OF THE GROUND STATE PHASE DIAGRAM

Our NQS calculations are performed on a torus of  $6 \times 6$  kagome unit cells ( $N = 3 \times 6 \times 6 = 108$  sites) with periodic boundary conditions along both lattice directions, complementing the cylindrical geometries used in our DMRG simulations. To represent the many-body wave function, we use a Vision Transformer (ViT) architecture with factored attention [68, 71, 72, 106], together with a variant described below. The two are employed in different regions of the phase diagram, according to the translation-symmetry content of the expected phases. We summarize here the main components of the original architecture and refer to Ref. [68] for a more detailed description. Both variants are illustrated in Fig. S5.

Given an input spin configuration  $\sigma = (\sigma_1^z, \dots, \sigma_N^z)$  in the computational ( $S^z$ ) basis, an NQS outputs the amplitude  $\Psi_\theta(\sigma) = \langle \sigma | \Psi_\theta \rangle \in \mathbb{C}$  of the variational state  $|\Psi_\theta\rangle$ , with  $\theta$  denoting the variational parameters. In the ViT of Ref. [68], the configuration is divided into  $n = N/P$  non-overlapping patches of  $P$  spins. Each patch configuration  $\mathbf{p}_i \in \{\pm 1\}^P$  is then mapped through a shared



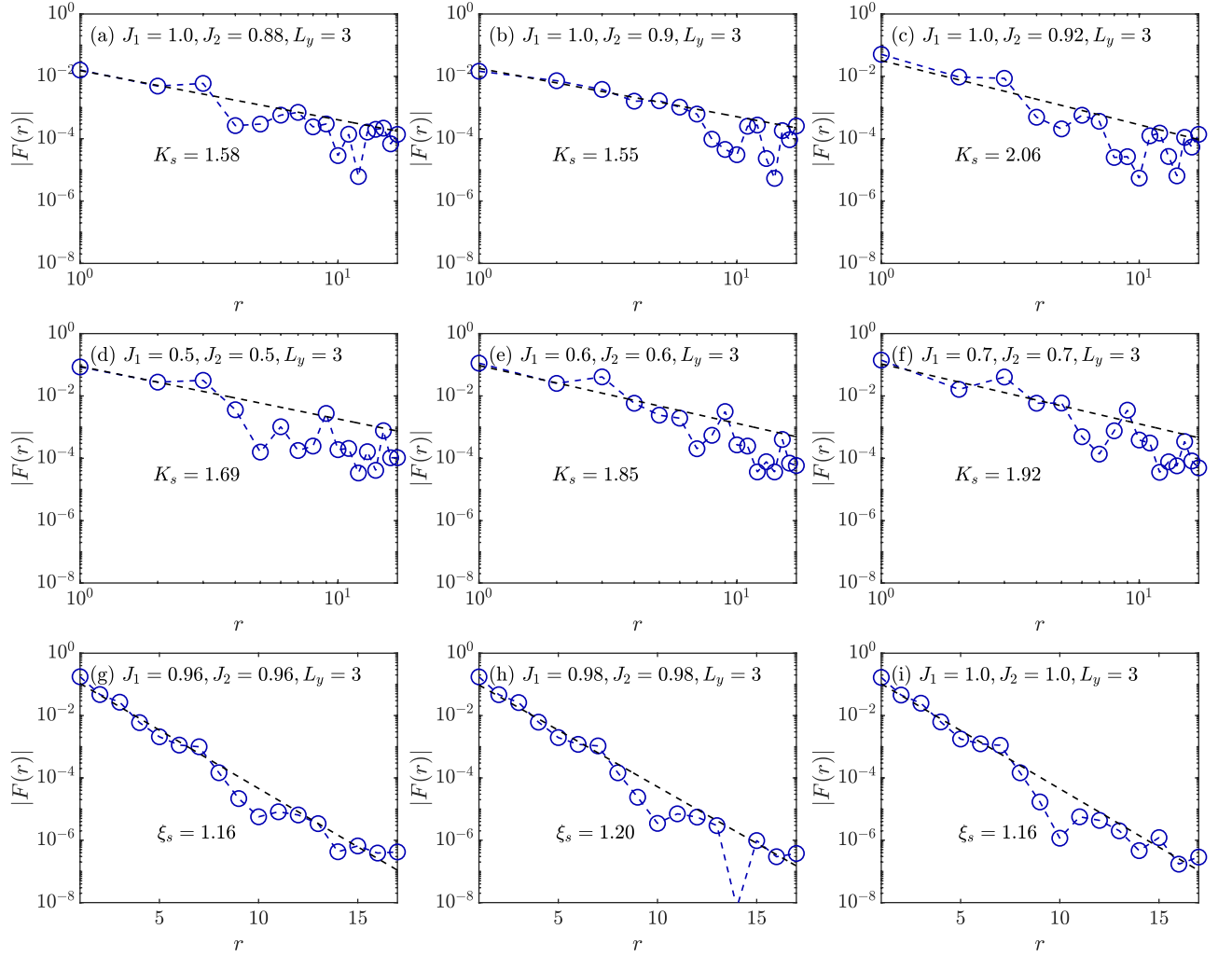


FIG. S3. **Spin correlations in the QSL and KSL phases.** (a-f) Double-logarithmic plots of the real-space spin correlation function  $F(r)$  in the QSL phase. The power exponents  $K_s$  are obtained by algebraic fitting with dash line. (g-i) Semi-logarithmic plots of  $F(r)$  in the KSL phase. The correlation lengths  $\xi_s$  are extracted by exponential fitting with dash line. All results are calculated on YC6 cylinders with  $L_x = 12$ , using a maximum bond dimension of  $D = 15000$ .

linear layer  $W$  to a  $d$ -dimensional embedding vector

$$\mathbf{u}_i^{(0)} = W(\mathbf{p}_i) \equiv E \mathbf{p}_i + \mathbf{b}, \quad (\text{S1})$$

where  $E \in \mathbb{R}^{d \times P}$  and  $\mathbf{b} \in \mathbb{R}^d$  are learnable. The resulting sequence of embedding vectors is passed through  $n_l$  transformer encoder layers implementing a factored attention mechanism with  $n_h$  heads [72, 106]. At each layer  $l$ , every head  $h = 1, \dots, n_h$  produces an attention vector

$$\mathbf{a}_i^h = \sum_{j=1}^n w_{ij}^h V^h(\mathbf{u}_j^{(l)}), \quad (\text{S2})$$

for each patch  $i$ , where  $V^h$  is a shared linear layer from  $\mathbb{R}^d$  to  $\mathbb{R}^{d/n_h}$  and  $w_{ij}^h$  is the factored attention weight, a learnable scalar depending only on the displacement between patches  $i$  and  $j$ , which renders the attention layer equivariant under patch-level translations of the input. The layer index is suppressed for all quantities other than the embedding vectors. The  $n_h$  attention vectors corresponding to patch  $i$  are then concatenated, linearly mixed, and passed through a residual feed-forward step to produce  $\mathbf{u}_i^{(l+1)}$  [68]. After the last encoder layer, the resulting vectors  $\mathbf{u}_i^{(n_l)}$  are summed over the patch index and mapped to  $\log \Psi_\theta(\sigma)$  by a complex-valued nonlinear layer [65, 68]. This final sum-pooling promotes the translation equivariance of the embedding-encoder stack to invariance of the variational state  $|\Psi_\theta\rangle$  under patch-level translations. We use this architecture for

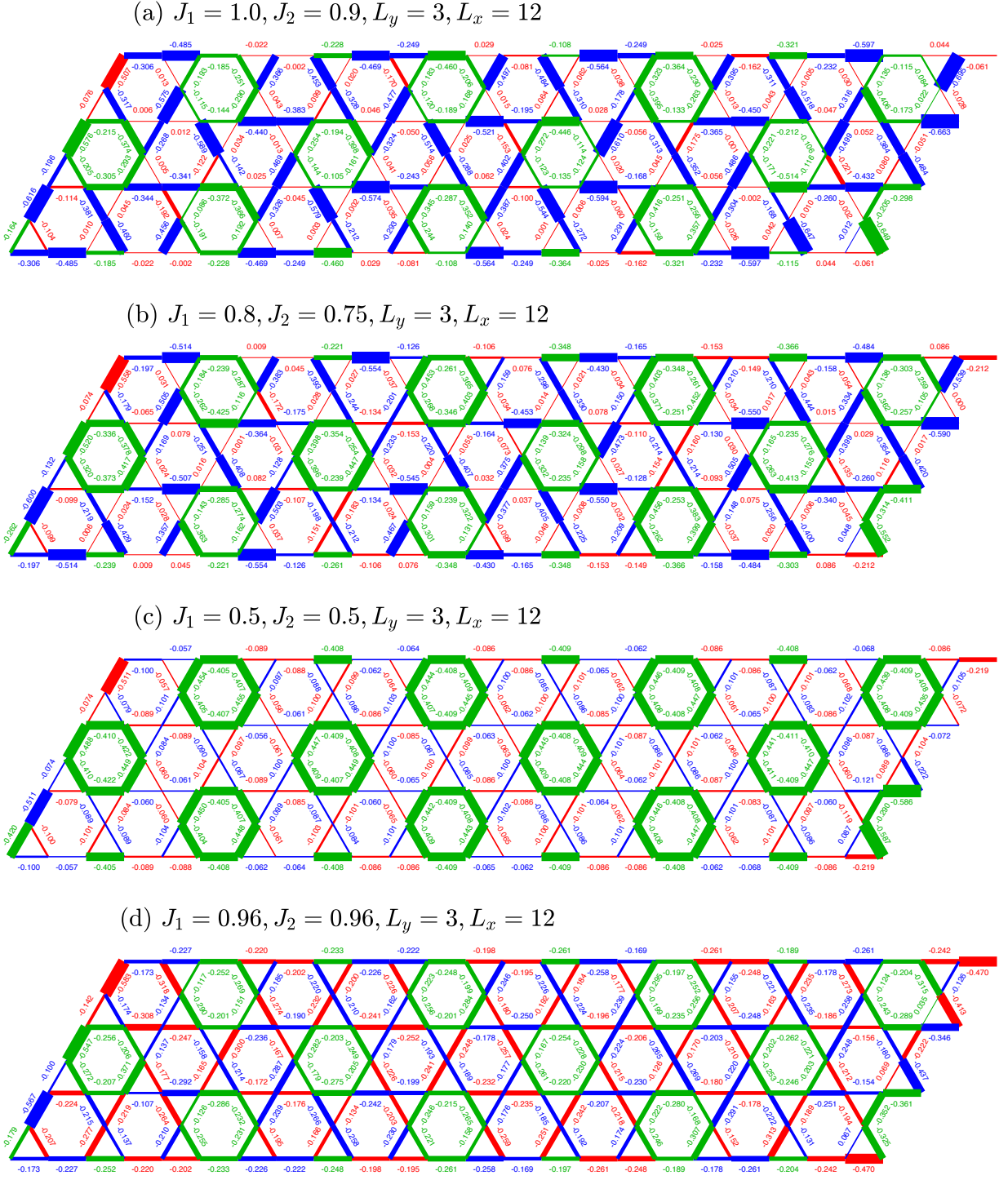


FIG. S4. **Bond energies in the QSL and KSL phases.** (a-c) Nearest-neighbor bond energies in the QSL phase for the three representative parameter sets, respectively. (d) Nearest-neighbor bond energies in the KSL phase. Different bond colors denote the distinct Heisenberg interactions defined in Eq. (1), while the bond thickness indicates the absolute value of the nearest-neighbor bond energy. All results are obtained on YC6 cylinders with  $L_x = 12$ , using a maximum bond dimension of  $D = 15000$ .

the scan of the line  $J_1 = 0.5$  in Fig. 3(b), choosing a patch of  $3 \times 3$  kagome lattice unit cells ( $P = 27$ ), which matches the magnetic unit cell of the  $\mathbf{Q} = (1/3, 1/3)$  ordered phase expected from the classical phase diagram [60] (see the inset of Fig. 1). This inductive bias is well suited to the magnetically ordered regions, while remaining compatible with disordered states that respect a finer translation symmetry. We use  $n_l = 4$ ,  $d = 160$  and  $n_h = 40$ , corresponding to approximately  $1.1 \times 10^6$  parameters.

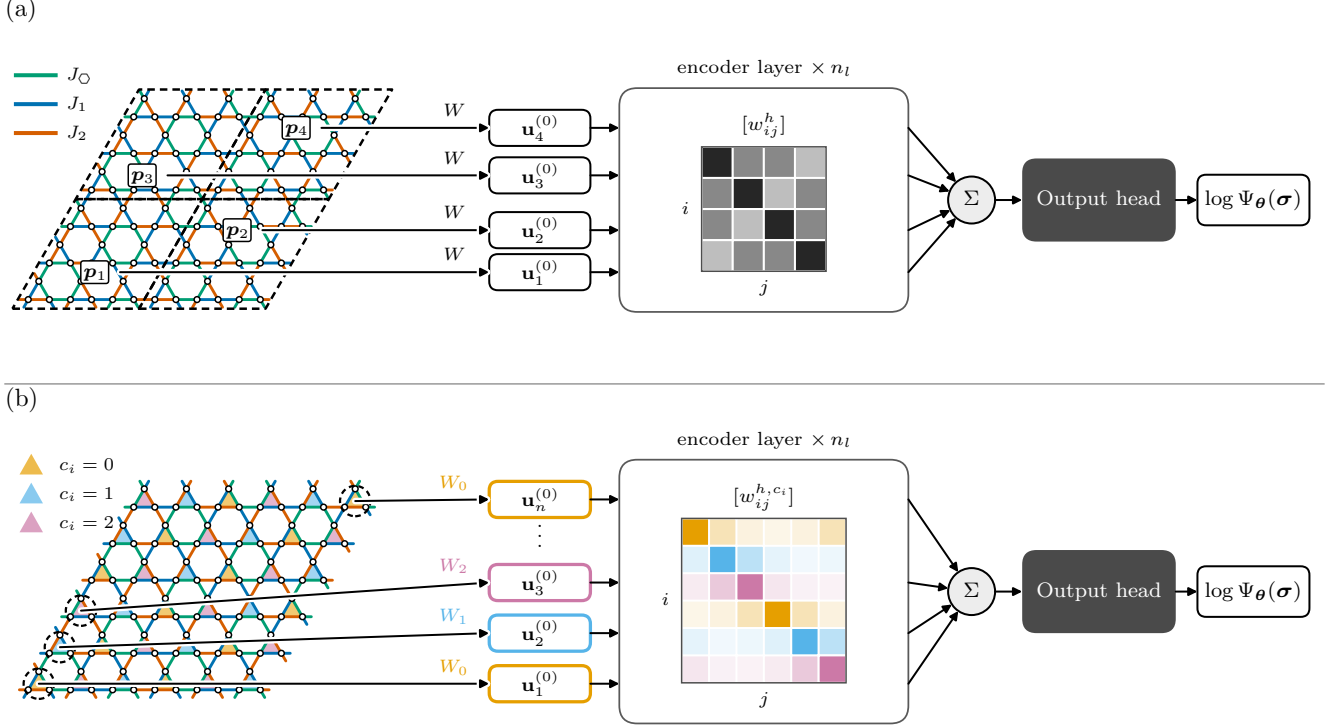


FIG. S5. **Schematic of the two ViT architecture variants.** (a) Architecture used on the  $J_1 = 0.5$  line. The spin configuration on the  $6 \times 6$  torus is divided into  $n = 4$  patches  $\mathbf{p}_i$  of  $3 \times 3$  kagome unit cells (shown with dashed outlines), each mapped by the shared linear layer  $W$  to an embedding vector  $\mathbf{u}_i^{(0)}$  [see Eq. (S1)]. The resulting embedding vectors pass through  $n_l$  encoder layers whose factored attention weights  $w_{ij}^h$ , Eq. (S2), depend only on the displacement between patches  $i$  and  $j$  (gray shading). The output vectors are summed over patches and mapped to  $\log \Psi_\theta(\sigma)$  by the output head. (b) Variant used on the  $J_1 = 1$  line. The patches are taken to be single kagome unit cells ( $n = 36$ ), colored by their class label  $c_i$ . The embedding layer  $W_{c_i}$  and the attention weights  $w_{ij}^{h,c_i}$  are shared only among patches of the same class, Eq. (S3).

The second architecture variant is used on the  $J_1 = 1$  line, close to the isotropic kagome limit ( $J_1 = J_2 = J_\square$ ), where the ground state is expected to respect the translation symmetry of the  $3J$  Hamiltonian. The patches are fixed to single kagome unit cells ( $P = 3$ ), with each patch  $i$  being assigned a class label  $c_i \in \{0, 1, 2\}$ , recording which of the three inequivalent kagome primitive cells of the nine-site  $3J$  unit cell it occupies. The embedding map and the attention weights of Eqs. (S1) and (S2) respectively are chosen to be class-dependent,

$$W(\mathbf{p}_i) \rightarrow W_{c_i}(\mathbf{p}_i), \quad w_{ij}^h \rightarrow w_{ij}^{h,c_i}, \quad (\text{S3})$$

replacing their shared counterparts. This change restricts the enforced translation symmetry from all kagome lattice translations, otherwise imposed by single unit cell patches, to only those of the  $3J$  Hamiltonian. In addition, given the finer patching, we adopt the spatial attenuation factor of Ref. [96], which multiplies the attention weights by a normalized exponential of the distance between patches with a learnable decay rate per layer and attention head. This introduces a locality inductive bias, which can be lifted or strengthened through optimization. We use  $n_l = 8$ ,  $d = 72$  and  $n_h = 12$ , corresponding to approximately  $4.4 \times 10^5$  parameters.

For every point  $(J_1, J_2)$  of the two scans, the ViT parameters  $\theta$  are optimized by minimizing the variational energy within the variational Monte Carlo framework [73]. Since the Hamiltonian (1) commutes with  $S_{\text{tot}}^z = \sum_i S_i^z$ , we perform Monte Carlo sampling within the  $S_{\text{tot}}^z = 0$  sector using update proposals that exchange pairs of opposite spins. We minimize the energy over  $10^4$  steps with  $M = 8192$  samples per step, using the Subsampled Projected-Increment Natural Gradient Descent (SPRING) optimization algorithm [107], which builds upon the stochastic reconfiguration (SR) [108, 109] and MinSR [110, 111] schemes, with momentum  $\mu = 0.9$  and a cosine-decaying [112, 113] learning rate from  $2 \times 10^{-2}$  to  $10^{-3}$ . The diagonal shift regularization is held constant at  $\lambda = 10^{-4}$  for the first variant, while for the second variant it is cosine decayed from  $10^{-3}$  to  $10^{-4}$  over the run, to stabilize the early optimization of the deeper network. After optimization, the static spin structure factor shown in Figs. 3 (b,e) and S2 are estimated with  $2^{16}$  Monte Carlo samples drawn from the final state.

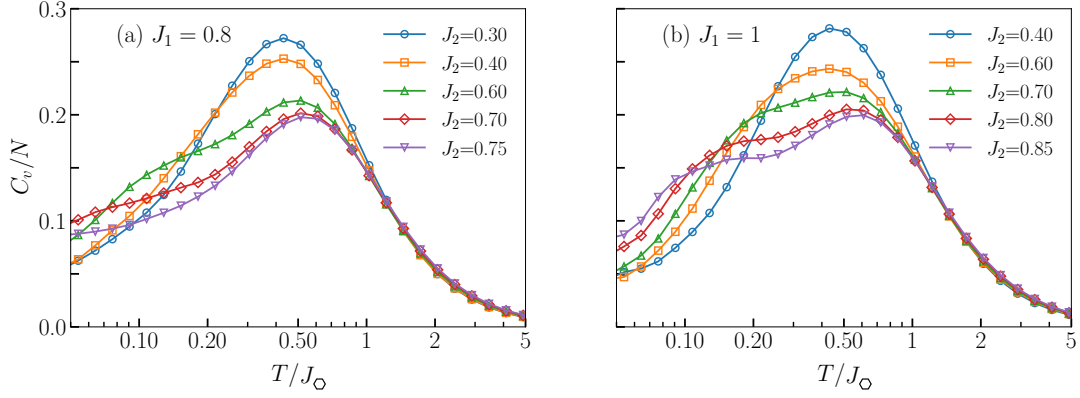


FIG. S6. **Specific heat of the 3J model from XTRG calculation** Specific heat at various  $J_2$  values at (a)  $J_1 = 0.8$  and (b)  $J_1 = 1$  with bond dimension  $D = 1000$ . The energy scale is set to  $J_O = 1$ .

### III. THERMAL TENSOR NETWORK CALCULATION OF THE SPECIFIC HEAT

#### A. exponential tensor renormalization group

The exponential tensor renormalization group (XTRG) method starts by constructing the high-temperature density operator  $\hat{\rho}(\tau) = e^{-\tau H}$ , where  $\tau$  denotes an infinitesimally small inverse temperature. This initial state can be reliably obtained via the Trotter–Suzuki decomposition [114, 115] or through series-expansion techniques [116]. A central ingredient of XTRG is its exponential evolution scheme: the density operator is repeatedly squared, i.e.,  $\hat{\rho}_{n+1} = \hat{\rho}_n \cdot \hat{\rho}_n$ , which doubles the inverse temperature at each step. This exponential growth drastically reduces the number of imaginary-time evolution and truncation steps, thereby enabling accurate simulations down to very low temperatures. In practice, we begin at a tiny inverse temperature  $\tau_0$  on the order of  $\sim 10^{-4}$  using the series-expansion method:

$$\hat{\rho}_0 \equiv \hat{\rho}(\tau_0) \simeq \sum_{k=0}^{N_c} \frac{(-\tau_0)^k}{k!} H^k,$$

with a truncated expansion order  $N_c = 4$ . We then apply the squaring procedure 19 times to access the lowest temperatures, enabling accurate temperature measurements down to  $T \sim 0.05 J_O$ .

In Fig. S6, we present the specific heat at several  $J_2$  values computed from XTRG with  $D = 1000$ , for the benefit of interested readers. The energy scale is set to  $J_O = 1$ . From the  $C_v(T)$  curves, distinct features of the ordered and QSL phases are clearly visible for both  $J_1 = 0.8$  and  $J_1 = 1.0$ . The ordered phase is characterized by a pronounced peak near  $T \approx 1$ , which gradually diminishes as the system enters the QSL phase. Concurrently, a shoulder-like feature emerges at  $T \approx 0.1$  within the QSL phase.

#### B. tangent-space tensor renormalization group

We also employ the tangent-space tensor renormalization group (tanTRG) method as another self-consistent computation at intermediate temperatures. The simulation procedure also begins by representing the spin Hamiltonian as an MPO, and the thermal density operator  $\rho(\beta) = e^{-\beta H}$  is initialized at an infinite-temperature limit (small inverse temperature  $\beta = 0$ ), providing an accurate starting point for the imaginary-time evolution. The cooling process to lower temperatures is then governed by the flow equation  $d\rho/d\beta = -H\rho$ . To maintain computational efficiency within a manageable bond dimension  $D$ , the evolution is restricted to the MPO manifold by projecting the right-hand side onto the tangent space at the current state, expressed as  $d\rho/d\beta = \mathcal{P}(-H\rho)$ , where  $\mathcal{P}$  is the tangent-space projector. Throughout the evolution, thermodynamic observables such as the free energy, internal energy, and specific heat are directly evaluated from the MPO representation of the density operator at each temperature step. Practically, we chose a we impose SU(2) symmetry with  $D = 1000 \sim 2000$  U(1) states retained, achieving a peak truncation error of roughly  $10^{-4}$  across the temperature range extending to  $T \sim 0.08$ .

Moreover, Fig. S7 compares XTRG and tanTRG calculations at  $(J_1, J_2) = (1.0, 0.9)J_O$  with bond dimensions increasing from 1000 to 1500 to 2000. The XTRG results converge well and show little dependence on bond dimension, whereas the tanTRG



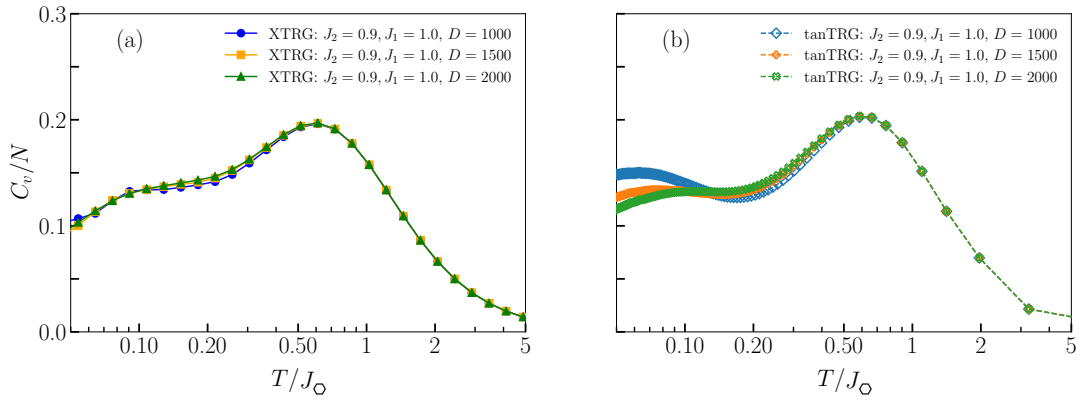


FIG. S7. **Specific heat of the 3J kagome models with various bond dimensions.** (a) Specific heat density from XTRG calculations with bond dimensions  $D = 1000$  (blue),  $1500$  (orange), and  $2000$  (green). (b) Corresponding tanTRG results with the same bond dimensions (blue, orange, green for  $D = 1000$ ,  $1500$ , and  $2000$ , respectively).

results exhibit a noticeable variation. Combined with Fig. 5, the tanTRG results gradually approach the XTRG ones, particularly at low temperatures ( $T/J_O < 0.5$ ). This behavior indicates that XTRG outperforms tanTRG in the low-temperature regime.